

1a(i)	No external forces act on the system of two objects. OR The two objects form a closed system.	B1
(ii)	Straight line from $(1.0, -3.0u)$ to $(2.0, 4.2u)$ Horizontal line at $p = 4.2u$ from $t = 2.0$ to 3.0 s	B1 B1
(iii)	Since the <u>total momentum before the collision is $6.0u$</u>, the <u>total momentum of the system remains constant at $6.0u$ at all times</u>. So final p of object B = $6.0u - 1.8u = 4.2u$ OR Since <u>p of object A decrease by $7.2u$ after the collision</u>, <u>p of particle B will increase by $7.2mu$</u>. So final p of particle B = $-3.0u + 7.2u = 4.2u$	B1

1b	$v_A = 1.8u / 9.0 = 0.2u$ $v_B = 4.2u / 1.0 = 4.2u$ relative speed of approach = $u_A - u_B = u - (-3u) = 4u$ relative speed of separation = $v_B - v_A = 4.2u - 0.2u = 4u$ Since the <u>relative speed of approach</u> and the <u>relative speed of separation</u> are <u>equal</u> , the collision is <u>elastic</u> . OR kinetic energy, $E_k = \frac{1}{2}mv^2 = \frac{1}{2}\frac{(mv)^2}{m} = \frac{p^2}{2m}$ before collision: $E_{k,before} = \frac{p_{A,before}^2}{2m_A} + \frac{p_{B,before}^2}{2m_B} = \frac{(9.0u)^2}{2(9.0)} + \frac{(-3.0u)^2}{2(1.0)} = 9.0u^2$ after collision: $E_{k,after} = \frac{p_{A,after}^2}{2m_A} + \frac{p_{B,after}^2}{2m_B} = \frac{(1.8u)^2}{2(9.0)} + \frac{(4.2u)^2}{2(1.0)} = 9.0u^2$ Since the <u>kinetic energy of the system before and after the collision remains the same</u> , the collision is <u>elastic</u> .	M1 M1 A1 (M1) (M1) (A1)
1c(i)	$F = \frac{\Delta p}{\Delta t}$ $= \frac{(1.8u - 9.0u)}{(2.0 - 1.0)}$ $= (-) 7.2u$ (accept + or – answer) Therefore, magnitude of F is 7.2u	M1 A0
(ii)	By Newton's second law, gradient of p vs t graph represents resultant force. During collision, <u>gradient</u> of graph of object A and object B have <u>equal magnitude</u> , showing that <u>forces</u> acting on A and B have <u>equal magnitude</u> . The gradients have <u>opposite signs</u> indicate that the two forces act <u>in opposite directions</u> . Hence the graphs are consistent with Newton's third law	B1 B1
Total:		10

2a	Total volume of molecules is negligible compared with volume occupied by the	P1
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